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Java SE 27 & JDK 27

Interface CompletionService<V>

Type Parameters:

`V` - the type of values the tasks of this service produce and consume

All Known Implementing Classes:

`ExecutorCompletionService`

`public interface` **`CompletionService<V>`**

A service that decouples the production of new asynchronous tasks from the consumption of the results of completed tasks. Producers `submit` tasks for execution. Consumers `take` completed tasks and process their results in the order they complete. A `CompletionService` can for example be used to manage asynchronous I/O, in which tasks that perform reads are submitted in one part of a program or system, and then acted upon in a different part of the program when the reads complete, possibly in a different order than they were requested.

Typically, a `CompletionService` relies on a separate `Executor` to actually execute the tasks, in which case the `CompletionService` only manages an internal completion queue. The `ExecutorCompletionService` class provides an implementation of this approach.

Memory consistency effects: Actions in a thread prior to submitting a task to a `CompletionService` *happen-before* actions taken by that task, which in turn *happen-before* actions following a successful return from the corresponding `take()`.

Since:

1.5

Method Summary

All Methods	Instance Methods	
Modifier and Type	**Method**	**Description**
`Future<V>`	**`poll`**`()`	Retrieves and removes the Future representing the next completed task, or `null` if none are present.
`Future<V>`	**`poll`**`(long timeout, TimeUnit unit)`	Retrieves and removes the Future representing the next completed task,

		waiting if necessary up to the specified wait time if none are yet present.
Future<V>	**submit**(Runnable task, V result)	Submits a Runnable task for execution and returns a Future representing that task.
Future<V>	**submit**(Callable<V> task)	Submits a value-returning task for execution and returns a Future representing the pending results of the task.
Future<V>	**take**()	Retrieves and removes the Future representing the next completed task, waiting if none are yet present.

Method Details

submit

`Future<V> submit(Callable<V> task)`

Submits a value-returning task for execution and returns a Future representing the pending results of the task. Upon completion, this task may be taken or polled.

Parameters:

`task` - the task to submit

Returns:

a Future representing pending completion of the task

Throws:

`RejectedExecutionException` - if the task cannot be scheduled for execution

`NullPointerException` - if the task is null

submit

```
Future<V> submit(Runnable task,
                 V result)
```

Submits a Runnable task for execution and returns a Future representing that task. Upon completion, this task may be taken or polled.

Parameters:

`task` - the task to submit

`result` - the result to return upon successful completion

Returns:

a Future representing pending completion of the task, and whose `get()` method will return the given result value upon completion

Throws:

`RejectedExecutionException` - if the task cannot be scheduled for execution

`NullPointerException` - if the task is null

take

```
Future<V> take()
        throws InterruptedException
```

Retrieves and removes the Future representing the next completed task, waiting if none are yet present.

Returns:

the Future representing the next completed task

Throws:

`InterruptedException` - if interrupted while waiting

poll

```
Future<V> poll()
```

Retrieves and removes the Future representing the next completed task, or `null` if none are present.

Returns:

the Future representing the next completed task, or `null` if none are present

poll

```
Future<V> poll(long timeout,
               TimeUnit unit)
        throws InterruptedException
```

Retrieves and removes the Future representing the next completed task, waiting if necessary up to the specified wait time if none are yet present.

Parameters:

`timeout` - how long to wait before giving up, in units of `unit`

`unit` - a `TimeUnit` determining how to interpret the `timeout` parameter

Returns:

the Future representing the next completed task or `null` if the specified waiting time elapses before one is present

Throws:

`InterruptedException` - if interrupted while waiting

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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